

Inferential Analysis on Study Habits and Academic Performances

A REPORT SUBMITTED FOR

Community Enhancement Program (CEP)

T.Y.B.Sc. Semester VI

SUBMITTED BY

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Declaration

We hereby declare that the report entitled **Inferential Analysis on Study Habits and Academic Performances** submitted as **Community Enhancement Program (CEP)** program report for T.Y.B.Sc. Semester VI for the academic year 2025 – 26 under NEP 2020, is our original work. This work has not formed the basis for any other course, previously of any degree, diploma of this college or any other college or any other University.

Place: Matunga, Mumbai.

Date:

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Certificate

This is to certify that the report entitled “**Inferential Analysis on Study Habits and Academic Performances** ” is the bonafide work carried out by following students

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This work is the part of **Community Enhancement Program (CEP)**

program for T.Y.B.Sc. Semester VI for the academic year 2025 – 26 conducted in November 2025.

This work has not formed the basis for any other course, previously for any degree, diploma of this college or any other college or any other University.

Ms. Shobhana Nadar

(Assistant Professor, Dept. of Mathematics & Statistics, Guru Nanak Khalsa College, Mumbai)

Place: Matunga, Mumbai.

Date:

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Study Habits & Academic Performance

PREFACE

The findings of this study highlight the importance of structured and consistent study habits in improving academic performance. Based on the statistical analysis conducted, it can be concluded that students who maintain regular study schedules and attend classes consistently tend to achieve higher CGPA. These results suggest that discipline and systematic planning in academic activities contribute more effectively to performance than irregular or unplanned efforts.

The study also indicates that maintaining a structured study routine plays a more significant role in academic success than merely increasing the number of study hours. In other words, the quality and consistency of study practices appear to be more influential than the quantity of time spent studying. Students who organize their learning through proper planning, time management, and routine-based study methods are more likely to perform better academically.

In addition, a strong association was observed between organized study habits and positive note-taking behaviour. Students who follow structured study schedules are more inclined to take regular and systematic notes, which may enhance understanding, retention, and revision efficiency. This reinforces the idea that effective study practices often work together and create a supportive academic environment for students.

On the other hand, the analysis revealed that stress level and gender do not have a statistically significant impact on CGPA in this dataset. Although minor numerical differences were observed, these differences were not strong enough to conclude that academic performance is directly influenced by stress level or gender.

Overall, the results of this research emphasize that consistent study planning, regular attendance, and organized academic behaviour are the most important factors influencing students' academic success. Developing disciplined study routines and maintaining engagement with classroom learning can significantly enhance overall academic performance.

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to the Department of Mathematics and Statistics, G.N. Khalsa College, Matunga, for providing us with the opportunity to undertake this project and apply our academic knowledge in a practical context. The support and academic environment of the department greatly contributed to the successful completion of this work.

We are especially thankful to Ms. Shobhna Nadar for her constant guidance, valuable suggestions, and encouragement throughout the project. Her clear explanations and continuous support helped us understand the proper application of statistical methods and the correct interpretation of results. Her guidance enabled us to approach the project with clarity and confidence.

We also appreciate the cooperation of all participants who contributed to the survey, making this analysis possible. Finally, we are grateful for the knowledge and experience gained through this project, which has enhanced our understanding of statistical research and strengthened our analytical skills.

Introduction

Academic performance is widely regarded as one of the most meaningful indicators of a student's learning, growth, and overall success within an educational system. It reflects not only the level of knowledge a student has acquired but also their ability to apply concepts, think critically, and perform consistently in assessments. In higher education especially, academic performance is often measured through the Cumulative Grade Point Average (CGPA), which serves as a standardized measure of achievement over time. Because CGPA can influence future academic opportunities, scholarships, and career prospects, understanding the factors that contribute to strong academic performance is of significant importance.

Among the many factors that may affect academic outcomes, study habits play a central role. Study habits refer to the regular practices and strategies students use to acquire and retain knowledge. These include the number of study hours dedicated each day, the consistency of a study routine, effective note-taking techniques, class attendance, time management, and preparation strategies before examinations. Positive and structured study habits are often associated with better comprehension, improved memory retention, and higher levels of academic achievement. On the other hand, irregular study patterns, procrastination, and lack of planning may negatively impact performance.

In addition to study habits, students frequently encounter challenges that can interfere with their academic progress. Stress, academic pressure, distractions from social media and technology, part-time work responsibilities, and personal issues can all influence concentration and productivity. These factors may reduce the effectiveness of even well-planned study efforts. Therefore, academic performance is not determined by intelligence alone, but by a combination of behavioral, psychological, and environmental influences.

Given the importance of academic success, it becomes essential to identify which specific study habits have the most significant impact on CGPA. By examining the relationship between study behaviors and academic results, this research aims to determine the key factors that contribute to improved academic performance. Understanding these relationships can provide valuable insights for students seeking to enhance their learning strategies.

The findings of this study are expected to benefit students by helping them recognize effective study practices and adopt more structured academic routines. Furthermore, the results may assist educators and institutions in guiding students toward better academic planning and support systems. Ultimately, by identifying and promoting productive study habits, this research seeks to contribute to improved learning outcomes and long-term academic success.

Data Description

In this research study, different variables have been selected to examine the relationship between study habits and academic performance among students. The purpose of defining these variables clearly is to ensure that the data collected can be properly analyzed and interpreted. The study includes both independent variables, which represent different study habit factors, and one dependent variable, which measures academic performance.

Gender is considered as one of the independent variables in this study. It is a categorical variable that represents whether a student is male or female. Since the categories do not have any numerical value or order, it is treated as a nominal variable. Stress level is another important variable included in the research. It represents the level of academic stress experienced by students and is classified into three categories: low, medium, and high. This variable helps in understanding whether stress has any noticeable effect on academic results.

Study hours refer to the amount of time a student spends studying each day. This variable is measured in ordered categories such as less than 1 hour, 1–2 hours, 2–3 hours, 3–4 hours, and more than 4 hours. Because these categories follow a logical sequence, study hours are treated as an ordinal variable. Study schedule is another factor considered in this research. It describes how consistently a student follows a study timetable and includes categories such as always, often, sometimes, rarely, and never. This helps in examining whether maintaining a regular schedule contributes to better academic outcomes.

Study routine represents the general study pattern followed by students and is treated as a categorical variable. Attendance is also included as an independent variable and indicates whether a student attends classes regularly or irregularly. Regular class attendance may provide better understanding of subjects and therefore influence academic performance. Note-taking habit is another variable that identifies whether a student practices note-taking during lectures or self-study. It is categorized as yes or no and helps determine whether this habit has a positive impact on learning. Study method refers to the preferred mode of studying, such as self-study, group study, or online study. This variable allows the research to compare different learning approaches and their effectiveness.

The dependent variable in this study is CGPA (Cumulative Grade Point Average), which measures overall academic performance. CGPA is treated as a continuous variable ranging from 0 to 10, as it can take decimal values within this range. Since it provides a precise measure of academic achievement, it serves as the main outcome variable of the research.

By clearly defining and classifying these variables, the study ensures that appropriate statistical methods can be applied for analysis. This structured approach helps in identifying which study habits significantly influence academic performance and contributes to a better understanding of effective learning practices among students.

Methodology

In this study, various statistical methods were used to analyze the data and draw meaningful conclusions about the relationship between study habits and academic performance. The data was first collected through a structured survey conducted among students. After collection, the responses were carefully checked for completeness and accuracy. The data was then cleaned to remove any missing or inconsistent entries and properly coded for analysis. All statistical analysis was carried out using R software, which is widely used for data analysis and statistical computing.

To begin the analysis, descriptive statistics were applied to summarize and understand the basic features of the data. Measures such as mean, median, standard deviation, and frequency distributions were calculated where appropriate. Graphical representations such as bar charts, histograms, and box plots were also used to visualize the distribution of CGPA and different study habit variables. This step helped in identifying general patterns, trends, and variations within the dataset.

Next, correlation analysis was performed to examine the strength and direction of relationships between selected variables. Correlation helps in understanding whether an increase or decrease in one variable is associated with a change in another variable. This provided preliminary insight into how certain study habits might be related to CGPA.

Further, group comparison tests were conducted depending on the type of variables involved. An independent sample t-test was used when comparing the mean CGPA between two groups, such as regular and irregular attendance. One-way ANOVA was applied when comparing mean CGPA across more than two groups, such as different stress levels or study methods. The chi-square test was used to examine associations between categorical variables. These tests helped determine whether observed differences between groups were statistically significant.

Finally, multiple linear regression analysis was used to identify which study habit factors significantly predict CGPA when considered together. This method allows for the examination of the combined effect of multiple independent variables on a continuous dependent variable. It also helps in determining which variables have the strongest influence on academic performance after controlling for other factors.

All statistical tests in this study were performed at a 5% level of significance. This means that results were considered statistically significant if the probability value (p-value) was less than 0.05, indicating that the findings were unlikely to have occurred by chance.

OBJECTIVES

The present study is conducted to examine how different study habits influence students' academic performance, measured through CGPA. Academic success is shaped by multiple behavioural and personal factors, and understanding these relationships can help in identifying effective learning practices. The objectives of this research are designed to systematically analyse differences, associations, and predictive effects between selected study habit variables and academic performance. The specific objectives of the study are as follows:

1. To evaluate differences in academic performance among students experiencing varying levels of academic stress.
2. To analyse the relationship between adherence to a study schedule and note-taking behaviour among students.
3. To examine whether there is a significant difference in mean CGPA between male and female students.
4. To assess the association between daily study duration and students' academic performance.
5. To compare the mean CGPA across students with different study routine patterns.
6. To identify the study habit variables that significantly predict students' overall academic performance (CGPA).

OBJECTIVE 1

To evaluate differences in academic performance among students experiencing varying levels of academic stress.

Why This Test?

→ We use One-Way ANOVA (Analysis of Variance).

Why ANOVA?

- Dependent variable (CGPA) → Continuous
- Independent variable (Stress Level) → Categorical (more than 2 groups)
- We want to compare means of more than two groups

Therefore, One-Way ANOVA is appropriate.

Model form:

$$CGPA_{ij} = \mu + \alpha_i + \epsilon_{ij}$$

Where:

- μ = overall mean CGPA
- α_i = effect of i th stress level
- ϵ_{ij} = random error

Hypothesis

Null Hypothesis (H_0): There is no significant difference in mean CGPA across stress levels.

$$\mu_1 = \mu_2 = \mu_3 = \mu_4$$

Alternative Hypothesis (H_1): At least one stress level has a different mean CGPA.

Level of Significance (LOS)

$$\alpha = 0.05(5\%)$$

R Code

```
# Convert stress variable to factor
dataset$StressLevel <- as.factor(dataset$Do.good.study.habits.help.reduce.exam.stress.)

# Run ANOVA
anova_model <- aov(CGPA ~ StressLevel, data = dataset)

summary(anova_model)
```

Output

```
# Convert stress variable to factor
> dataset$StressLevel <- as.factor(dataset$Do.good.study.habits.help.reduce.exam.stress.)
> # Run ANOVA
> anova_model <- aov(CGPA ~ StressLevel, data = dataset)
> summary(anova_model)
      Df Sum Sq Mean Sq F value Pr(>F)
StressLevel 3  4.62  1.540  0.996 0.398
Residuals 96 148.47  1.546
```

Decision Criteria

- If $p\text{-value} \leq 0.05 \rightarrow \text{Reject } H_0$
- If $p\text{-value} > 0.05 \rightarrow \text{Fail to Reject } H_0$

Here,

$$p = 0.398$$

Since

$$0.398 > 0.05$$

We fail to reject H_0 .

Interpretation

The ANOVA test shows that the p-value (0.398) is greater than the 5% level of significance.

This means that the differences in mean CGPA across stress levels are not statistically significant.

Although there may be small numerical differences in CGPA means, these differences are likely due to random variation and not due to stress level.

Conclusion

At 5% level of significance, there is no significant difference in mean CGPA across different stress levels.

Hence, stress level does not appear to significantly affect CGPA in this dataset.

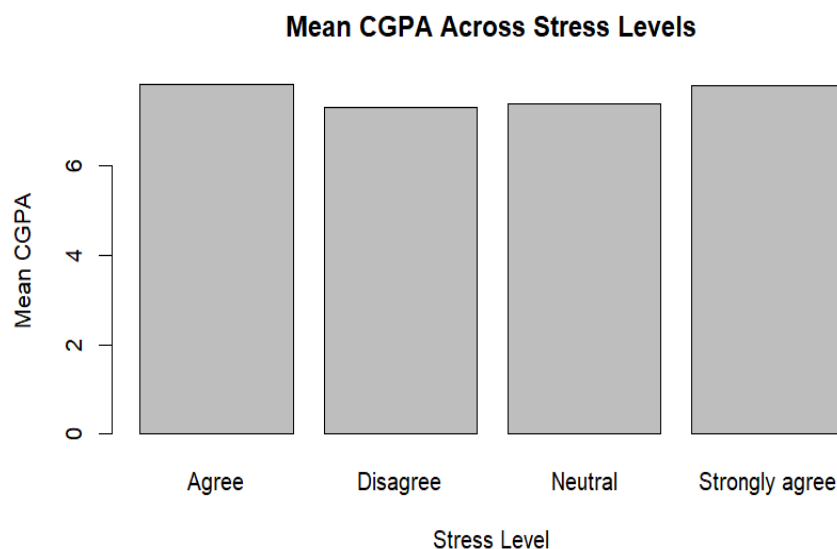
Visual representation

Code

```
# Calculate mean CGPA for each stress level
mean_cgpa <- tapply(dataset$CGPA, dataset$StressLevel, mean, na.rm = TRUE)

# Simple bar plot
barplot(mean_cgpa,
        xlab = "Stress Level",
        ylab = "Mean CGPA",
        main = "Mean CGPA Across Stress Levels")
```

Output



Interpretation

1. The bar graph shows that the mean CGPA is almost similar across all stress levels.
2. Students who strongly agree that good study habits reduce stress have slightly higher CGPA compared to others.
3. Those who disagree have marginally lower CGPA, but the difference is very small. The variation among groups is minimal, indicating no major performance gap.
4. This visual observation supports the ANOVA result ($p = 0.398$), which shows no statistically significant difference.
5. Hence, stress level does not appear to significantly affect CGPA in this dataset.

OBJECTIVE 2

To examine whether there is a significant association between Study Schedule and Note Taking habits among students.

Why This Test?

We use the Chi-Square Test of Independence.

Why Chi-Square Test?

- StudySchedule → Categorical variable
- NoteTaking → Categorical variable
- We want to check whether the two categorical variables are associated.

Hence, Chi-square test is appropriate.

Hypothesis

Null Hypothesis (H_0):

There is no association between Study Schedule and Note Taking habits.
(Both variables are independent.)

Alternative Hypothesis (H_1):

There is a significant association between Study Schedule and Note Taking habits.

Level of Significance (LOS)

$$\alpha = 0.05(5\%)$$

R Code

```
table_data <- table(dataset$StudySchedule,  
                    dataset$Do.you.take.your.own.notes.while.studying.)  
  
table_data  
chisq.test(table_data)
```

Output

```
table_data <- table(dataset$StudySchedule,
+                   dataset$Do.you.take.your.own.notes.while.studying.)
> table_data
```

	No	Rarely	Sometimes	Yes, always
Always	0	1	0	7
Never	2	4	2	5
Often	0	7	9	8
Rarely	0	5	13	5
Sometimes	0	3	15	14

```
> chisq.test(table_data)
```

Pearson's Chi-squared test

data: table_data
X-squared = 31.491, df = 12, p-value = 0.001657

Decision Criteria

- If $p\text{-value} \leq 0.05 \rightarrow \text{Reject } H_0$
- If $p\text{-value} > 0.05 \rightarrow \text{Fail to Reject } H_0$

Here,

$$0.001657 < 0.05$$

So, we reject the null hypothesis.

Interpretation

Since the p-value is less than 0.05, there is a statistically significant association between Study Schedule and Note Taking habits.

This means students who follow a structured study schedule (Always/Often) are more likely to take notes regularly, whereas those who never follow a schedule show less consistent note-taking behavior.

The two variables are not independent.

Conclusion

At the 5% level of significance, we conclude that Study Schedule and Note Taking habits are significantly associated.

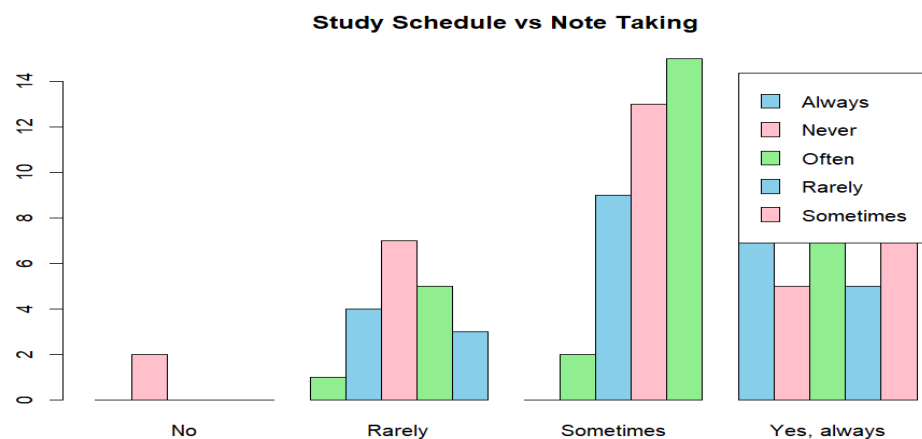
This suggests that maintaining a proper study schedule may influence students' note-taking behavior.

Visual representation

Code

```
barplot(table_data,
        beside = TRUE,
        col = c("skyblue", "pink", "lightgreen"),
        legend = TRUE,
        main = "Study Schedule vs Note Taking")
```

Output



Interpretation

1. The bar chart shows a clear relationship between study schedule and note-taking habits.
2. Students who often or sometimes follow a study schedule tend to report “Yes, always” or “Sometimes” for note taking in higher numbers.
3. Those who never follow a schedule show comparatively lower consistent note-taking behavior.
4. Very few students who do not follow a schedule report regular note taking.
5. The pattern suggests that a structured study schedule is associated with more active note-taking.
6. This visual trend supports the Chi-square result ($p = 0.001657$), indicating a significant association between the two variables.

OBJECTIVE 3

To examine whether there is a significant difference in mean CGPA between male and female students.

Why This Test?

We use the Independent Sample t-test (Welch Two Sample t-test) because:

- CGPA → Continuous variable
- Gender → Categorical variable (2 groups: Male & Female)
- We want to compare the means of two independent groups

Hence, the t-test is appropriate.

Hypothesis

Null Hypothesis (H_0):

There is no significant difference in mean CGPA between male and female students.

$$\mu_{Female} = \mu_{Male}$$

Alternative Hypothesis (H_1):

There is a significant difference in mean CGPA between male and female students.

$$\mu_{Female} \neq \mu_{Male}$$

Level of Significance (LOS)

$$\alpha = 0.05(5\%)$$

Code

```
dataset$Gender <- as.factor(dataset$Gender)
t.test(CGPA ~ Gender, data = dataset)
```

Output

```
> dataset$Gender <- as.factor(dataset$Gender)
> t.test(CGPA ~ Gender, data = dataset)
```

Welch Two Sample t-test

```
data: CGPA by Gender
t = -1.1618, df = 81.192, p-value = 0.2487
```

alternative hypothesis: true difference in means between group Female and group Male is not equal to 0
95 percent confidence interval:
-0.8069472 0.2119573
sample estimates:
mean in group Female mean in group Male
7.516727 7.814222

Decision Criteria

- If $p\text{-value} \leq 0.05 \rightarrow \text{Reject } H_0$
- If $p\text{-value} > 0.05 \rightarrow \text{Fail to Reject } H_0$

Here,

$$0.2487 > 0.05$$

So, we fail to reject the null hypothesis.

Interpretation

Although male students have a slightly higher mean CGPA (7.81) compared to female students (7.52), the difference is not statistically significant.

The p-value (0.2487) is greater than 0.05, indicating that the observed difference may be due to random variation.

The boxplot also shows overlapping distributions, supporting the lack of significant difference.

Conclusion

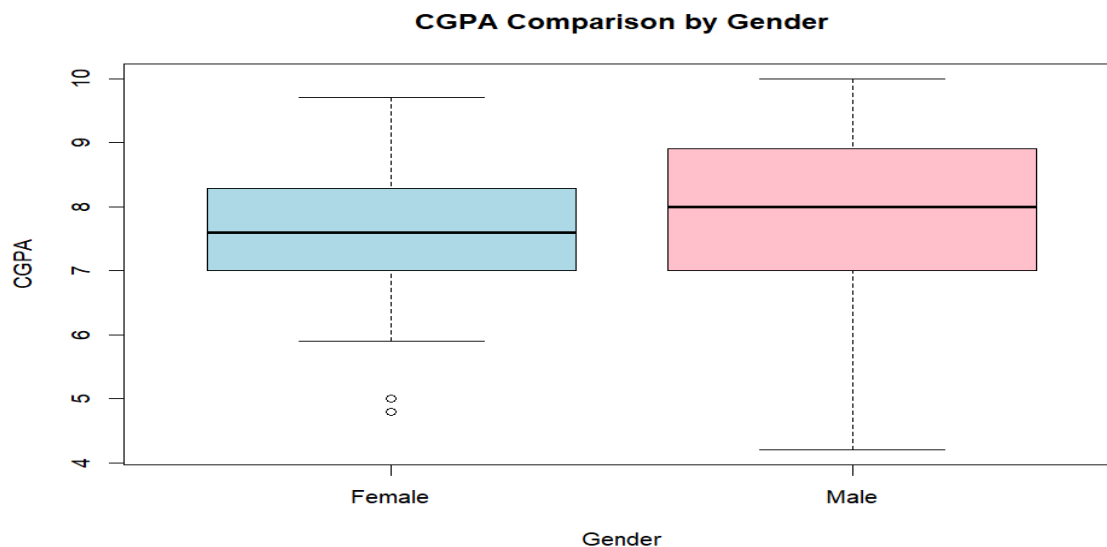
At the 5% level of significance, there is no significant difference in academic performance (CGPA) between male and female students in this dataset.

Visual representation

Code

```
boxplot(CGPA ~ Gender,
  data = dataset,
  main = "CGPA Comparison by Gender",
  xlab = "Gender",
  ylab = "CGPA",
  col = c("lightblue", "pink"))
```

Output



Interpretation

1. The boxplot shows that the median CGPA of male students is slightly higher than that of female students.
2. However, the boxes overlap considerably, indicating that the overall performance distribution is quite similar for both genders.
3. The spread (interquartile range) appears slightly wider for males, suggesting more variability in their CGPA scores.
4. Female students show a few lower outliers, indicating some unusually low CGPA values.
5. Overall, the distributions are similar, and there is no strong visual evidence of a major difference in academic performance between genders.

Objective 4

To assess the association between daily study duration and students' academic performance.

Formulation of Hypothesis

H_0 : There is no significant relationship between daily study hours and academic performance (CGPA) among students.

H_1 : There is a significant relationship between daily study hours and academic performance (CGPA) among students

Level of Significance,

$$\alpha = 0.05$$

Code

```
> #To examine the relationship between study hours and academic performance (CGPA)
> cgpa<-dataset$Enter.your.overall.CGPA.in.the.examination
> study_hours<-dataset$How.many.hours.do.you.study.daily..on.average..
> install.packages("dplyr")
> library(dplyr)
> unique(study_hours)
[1] "Less than 1" "1\u00962" "2\u00963" "3\u00964" "More than 4"
> study_hours_clean<-trimws(tolower(study_hours))
> study_hours_num <- recode(study_hours,
+                           "Less than 1" = 1,
+                           "1-2" = 2,
+                           "2-3" = 3,
+                           "3-4" = 4,
+                           "More than 4" = 5)
> study_hours_num <- ifelse(grepl("less", study_hours_clean), 1,
+                           ifelse(grepl("1", study_hours_clean), 2,
+                           ifelse(grepl("2", study_hours_clean), 3
+                           ,
+                           ifelse(grepl("3", study_hours_clean), 4,
+                           ifelse(grepl("more", study_hours_clean), 5, NA))))
> table(study_hours_clean,useNA = "ifany")
study_hours_clean
  1\u00962    2\u00963    3\u00964 less than 1 more than 4
      38         23         16         10         14
> cgpa<-as.numeric(cgpa)
> cor.test(study_hours_num, cgpa, method = "spearman")
```

Spearman's rank correlation rho

data: study_hours_num and cgpa

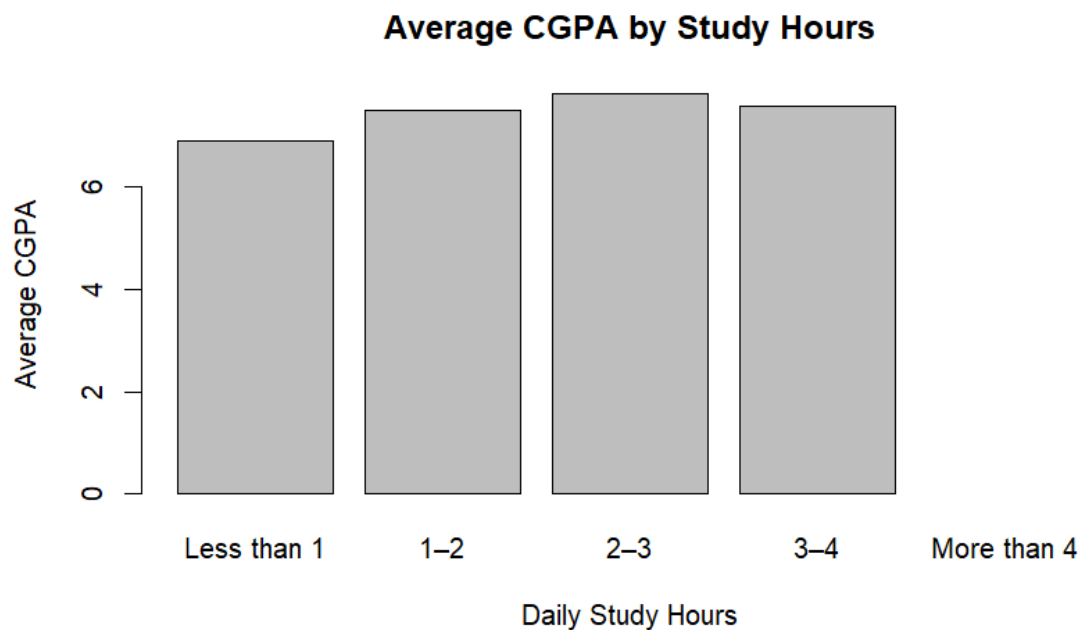
S = 130356, p-value = 0.0295

alternative hypothesis: true rho is not equal to 0

```
sample estimates:
      rho
0.2177859
```

Step 4:- visual representation

```
> labels <- c("Less than 1","1-2","2-3","3-4","More than 4")
> barplot(tapply(cgpa, study_hours_num, mean),
+         names.arg = labels,
+         xlab = "Daily Study Hours",
+         ylab = "Average CGPA",
+         main = "Average CGPA by Study Hours")
```



Decision Criteria

If $p\text{-value} < 0.05$, we reject H_0

Interpretation and Conclusion

$p\text{-value} = 0.0295$

$0.0295 < 0.05$, we reject H_0

$\rho = 0.2177859$

which is positive means as study hours increases , CGPA trends to increase

The study concludes that daily study hours have a significant influence on academic performance at 5% LOS. Although the relationship is weak, increasing study duration is associated with improvement in students' academic achievement.

Objective 5

To determine whether there is a significant difference in mean CGPA among different study routine groups.

One way Anova

Formulation of Hypothesis

H₀: $\mu_1 = \mu_2 = \mu_3 = \dots$ (All study routine groups have equal mean CGPA)

H₁: At least one group mean is different

Level of Significance, $\alpha = 0.05$

Code

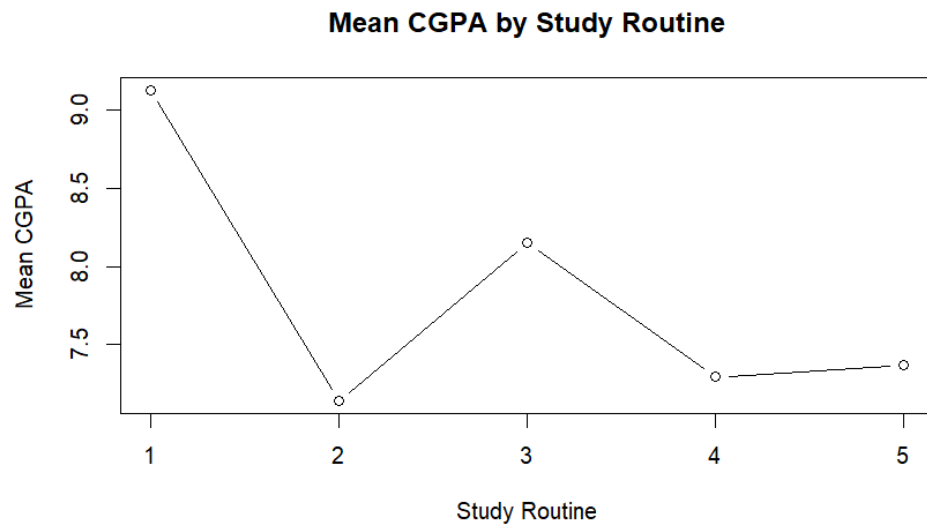
```
> #To determine whether there is a significant difference in mean CGPA among different study routine groups.
> # Convert StudySchedule into factor
> dataset$StudySchedule <- as.factor(dataset$StudySchedule)
> # Remove missing values
> dataset <- na.omit(dataset)
> # Run One-Way ANOVA
> anova_model <- aov(CGPA ~ StudySchedule, data = dataset)
> # View summary
> summary(anova_model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
StudySchedule	4	32.18	8.044	6.321	0.000149 ***
Residuals	95	120.91	1.273		

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Visual representation

```
> plot(means <- tapply(dataset$CGPA, dataset$StudySchedule, mean),
+      type = "b",
+      xlab = "Study Routine",
+      ylab = "Mean CGPA",
+      main = "Mean CGPA by Study Routine")
```



Decision Criteria

If

$p\text{-value} < 0.05$, we reject H_0

Interpretation and Conclusion

$p\text{-value} = 0.000149$

$0.000149 < 0.05$, we reject H_0 and conclude that

There is a significant difference in mean CGPA among study routine groups at 5% LOS

Objective 6

To identify the study habit variables that significantly predict students' overall academic performance (CGPA).

Multiple regression model

- $CGPA = \beta_0 + \beta_1(StudyHours) + \beta_2(StudyMethod) + \beta_3(StudySchedule) + \beta_4(Attendance) + \beta_5(StudyConsistently) + \beta_6(ConceptClear) + \epsilon$

Where:

CGPA = Dependent variable

StudyHours, StudyMethod, StudySchedule, Attendance = Independent variables

ϵ = Error term

- This model examines how different study habit factors influence students' CGPA.

Formulation of Hypothesis

Overall Hypothesis

H₀ :All regression coefficients are equal to zero.

$$H_0: \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = \beta_6 = 0$$

H₁ :At least one regression coefficient is not zero.

$$H_1: \text{At least one } \beta_i \neq 0$$

Individual Hypothesis

For Study Schedule

H₀: $\beta_3 = 0$ (Study schedule has no effect on CGPA)

H₁: $\beta_3 \neq 0$ (Study schedule affects CGPA)

Similarly for:

- Study Hours
- Study Method
- Attendance
- Study Schedule
- Study consistency
- Concept clearance

Level of Significance

$$\alpha = 0.05$$

Code

```
> dataset <- read.csv("C:/Users/Swapnali/OneDrive/Desktop/Study Habits and  
Academic Performance Questionnaire (Responses).csv",fileEncoding = "latin  
1")  
> view(dataset)
```

```

> names(dataset)[names(dataset) ==
+ "How.many.hours.do.you.study.daily..on.average.."] <- "
StudyHours"
> names(dataset)[names(dataset) ==
+ "I.revise.topics.regularly.after.they.are.taught.in.cla
ss." ] <- "Revision"
> names(dataset)[names(dataset) ==
+ "which.study.method.do.you.prefer.the.most." ] <- "Study
Method"
> names(dataset)[names(dataset) ==
+ "Do.you.follow.a.fixed.study.schedule." ] <- "StudySched
ule"
> names(dataset)[names(dataset) ==
+ "I.attend.classes.or.lectures.regularly." ] <- "Attendan
ce"
> names(dataset)[names(dataset) ==
+ "I.practice.solving.problems.or.questions.related.to.my
.subjects." ] <- "practice_paper"
> names(dataset)[names(dataset) ==
+ "I.study.consistently.rather.than.only.before.examinati
ons." ] <- "study_consistently"
> names(dataset)[names(dataset) ==
+ "My.study.habits.help.me.understand.concepts.clearly." ]
<- "concept_clear"
> names(dataset)[names(dataset) ==
+ "Enter.your.overall.CGPA.in.the.examination" ] <- "CGPA"
> dataset$StudyMethod <- as.factor(dataset$StudyMethod)
> dataset$StudySchedule <- as.factor(dataset$StudySchedule)
> dataset$Attendance <- as.factor(dataset$Attendance)
> dataset$Gender <- as.factor(dataset$Gender)
> dataset$Age.Group <- as.factor(dataset$Age.Group)
> str(dataset$CGPA)
chr [1:101] "7" "7" "8.6" "8.5" "9.8" "7.5" "8.9" "9.51" "7.6" "9.27" "8.
2" "8.15" "6.05" ...
> dataset$CGPA <- as.numeric(dataset$CGPA)

```

Warning message:

NAs introduced by coercion

```

> summary(dataset$CGPA)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
 4.200  7.000  7.855  7.651  8.600 10.000     1
> sum(is.na(dataset$CGPA))
[1] 1
> mlr_model <- lm(CGPA ~ StudyHours + StudyMethod +
+ StudySchedule + study_consistently+concept_clear+Attendanc
e,
+ data = dataset)
> summary(mlr_model)

```

Call:

```
lm(formula = CGPA ~ StudyHours + StudyMethod + StudySchedule +
study_consistently + concept_clear + Attendance, data = dataset)
```

Residuals:

```

      Min       1Q   Median       3Q      Max
-2.74140 -0.71341  0.06081  0.71751  2.33681

```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	8.401e+00	8.243e-01	10.192	< 2e-16	***
StudyHours2\u00963	1.966e-01	3.142e-01	0.626	0.53326	
StudyHours3\u00964	1.359e-01	3.489e-01	0.390	0.69782	
StudyHoursLess than 1	-3.744e-01	4.156e-01	-0.901	0.37004	
StudyHoursMore than 4	3.929e-01	3.790e-01	1.037	0.30273	
StudyMethodVirtual	1.358e-01	2.783e-01	0.488	0.62686	
StudyScheduleNever	-1.542e+00	5.892e-01	-2.617	0.01046	*
StudyScheduleOften	-6.610e-01	5.216e-01	-1.267	0.20846	
StudyScheduleRarely	-1.425e+00	5.323e-01	-2.677	0.00888	**
StudyScheduleSometimes	-1.563e+00	4.975e-01	-3.142	0.00230	**
study_consistently	-3.804e-02	1.253e-01	-0.304	0.76210	
concept_clear	-9.183e-05	1.351e-01	-0.001	0.99946	
AttendanceYes	6.248e-01	2.905e-01	2.151	0.03428	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.11 on 87 degrees of freedom

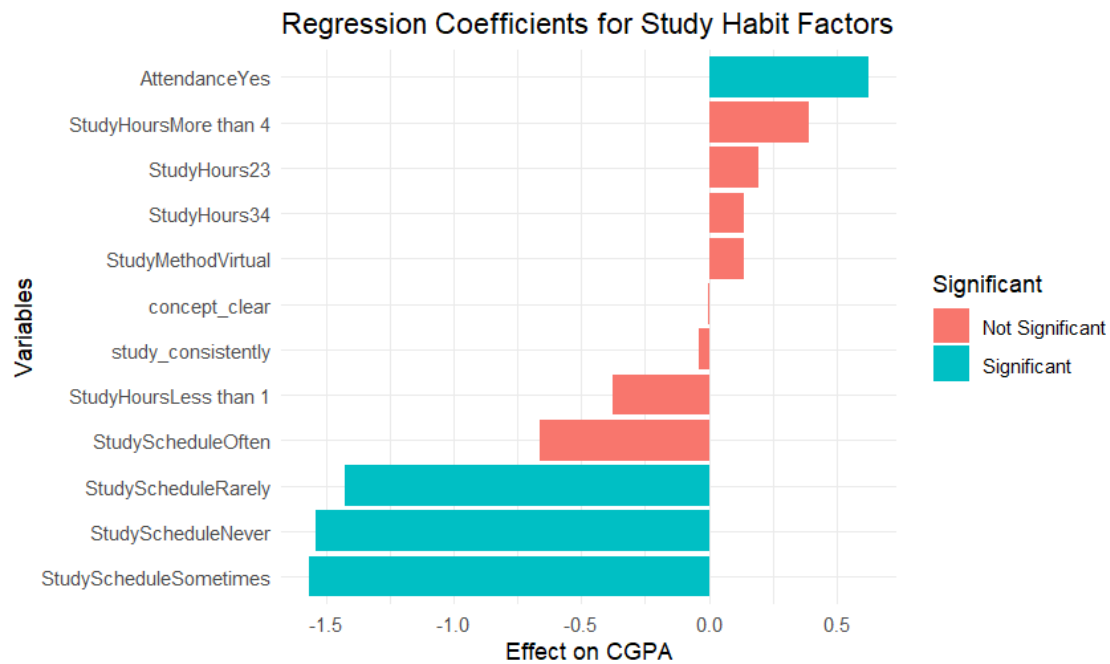
(1 observation deleted due to missingness)

Multiple R-squared: 0.3002, Adjusted R-squared: 0.2036

F-statistic: 3.109 on 12 and 87 DF, p-value: 0.001056

Step 4:- Visual representation

```
> install.packages("ggplot2")
> library(ggplot2)
> # Extract coefficients
> coef_df <- as.data.frame(summary(mlr_model)$coefficients)
> coef_df$Variable <- rownames(coef_df)
> # Remove intercept
> coef_df <- coef_df[-1, ]
> # Create significance label
> coef_df$Significant <- ifelse(coef_df$`Pr(>|t|)` < 0.05,
+                               "Significant", "Not Significant")
> # Plot
> ggplot(coef_df, aes(x = reorder(Variable, Estimate),
+                         y = Estimate,
+                         fill = Significant)) +
+   geom_bar(stat = "identity") +
+   coord_flip() +
+   labs(title = "Regression Coefficients for Study Habit Factors",
+         x = "Variables",
+         y = "Effect on CGPA") +
+   theme_minimal()
```



Decision Criteria

If

$p\text{-value} < 0.05$, we reject H_0

Overall interpretation and conclusion

$p\text{-value} = 0.001056$

$0.001056 < 0.05$, we reject H_0 and conclude that

At 5% level of significance, the Multiple Regression Model indicates that study habit factors collectively have a significant effect on academic performance. The model explains approximately 29.9% of the variation in CGPA.

FUTURE PREDICTION

Based on the findings of this study, it is expected that students who adopt disciplined and well-planned study routines will achieve better academic performance in the future. The results clearly indicate that structured academic behaviour plays a central role in determining overall success. Therefore, encouraging students to develop systematic and consistent study practices may lead to improved learning outcomes over time.

The study suggests that strengthening time-management skills and maintaining organised study habits, such as following a regular study schedule and practicing effective note-taking, can significantly enhance learning effectiveness. These habits promote better understanding, retention of concepts, and efficient revision, which ultimately contribute to higher academic achievement. Rather than focusing only on increasing study hours, students should emphasize quality, consistency, and proper planning in their academic preparation.

Furthermore, regular attendance and continuous engagement in classroom activities are expected to positively influence CGPA in future cohorts. Attending lectures consistently provides students with direct exposure to subject content, clarifies doubts, and supports active participation in the learning process.

Overall, the findings indicate that strengthening disciplined and structured study habits will be the most influential factor in improving academic performance among students in the future. By promoting organized learning practices and consistent academic involvement, institutions and students alike can work toward achieving sustained academic success.

CONCLUSION

The findings of this study highlight the importance of structured and consistent study habits in improving academic performance. Based on the statistical analysis conducted, it can be concluded that students who maintain regular study schedules and attend classes consistently tend to achieve higher CGPA. These results suggest that discipline and systematic planning in academic activities contribute more effectively to performance than irregular or unplanned efforts.

The study also indicates that maintaining a structured study routine plays a more significant role in academic success than merely increasing the number of study hours. In other words, the quality and consistency of study practices appear to be more influential than the quantity of time spent studying. Students who organize their learning through proper planning, time management, and routine-based study methods are more likely to perform better academically.

In addition, a strong association was observed between organized study habits and positive note-taking behaviour. Students who follow structured study schedules are more inclined to take regular and systematic notes, which may enhance understanding, retention, and revision efficiency. This reinforces the idea that effective study practices often work together and create a supportive academic environment for students.

On the other hand, the analysis revealed that stress level and gender do not have a statistically significant impact on CGPA in this dataset. Although minor numerical differences were observed, these differences were not strong enough to conclude that academic performance is directly influenced by stress level or gender.

Overall, the results of this research emphasize that consistent study planning, regular attendance, and organized academic behaviour are the most important factors influencing students' academic success. Developing disciplined study routines and maintaining engagement with classroom learning can significantly enhance overall academic performance.

REFERENCE

- Student Survey Data (Primary Data)
- Montgomery, D. C. — Introduction to Linear Regression Analysis
- Field, A. — Discovering Statistics Using R
- R Statistical Software Documentation